GEBZE TECHNICAL UNIVERSITY PHYSICS DEPARTMENT

PHYS 424 : CLASSICAL MECHANICS II

2025 Spring Semester, Homework III (Thursday, May 26, 2025)

INSTRUCTIONS :

1. You cannot ask someone to show you how to completely solve the problem, nor can you accept such extensive help.
2. Under no circumstances should you copy or even look at someone's solutions.
3. Write down your solutions clearly and readable. Submit them on the due date during the lecture.

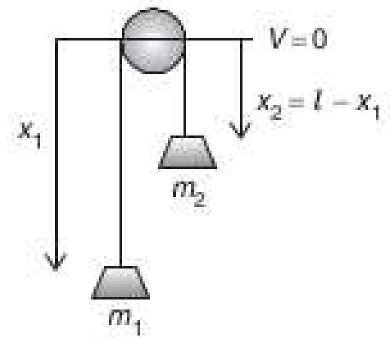
---

1) In the following figure you see a Atwood's machine. The masses are suspended by a massless, inextensible string over a frictionless, massless pulley.

a) What is the constraint of the motion ?

b) Evaluate the Lagrangian, and obtain the equation of motion.

c) Find the Hamiltonian of the system.



2) Consider an attractive central force with the form $F(r) = -\dfrac{k}{r^n}$.

a) Find the potential energy $U(r)$ for such a force.

b) Write the effective potential function, or energy, $V(r)$.

c) Show that a stable circular orbit exists only for $n < 3$.

3) Explain Kepler's laws with their mathematical expressions.

4) A particle of mass m is bound by a linear potential $U = kr$.

a) For what energy and angular momentum will the orbit be a circle of radius r about the origin ?

b) What is the angular frequency of this circular motion ?

c) What is the effective potential ?

5) A uniform rod of length b stands vertically upright on a rough floor and then tips over. What is the rod's angular velocity when it hits the floor ?

6) Solve the following problems.

a) A three-particle system of masses m_i and coordinates (x_1, x_2, x_3) as follows:

$$m_1 = 3m,\ (b, 0, b)\,; \quad m_2 = 4m,\ (b, b, -b)\,; \quad m_3 = 2m,\ (-b, b, 0)\,.$$

Find the elements of moment of inertia tensor for this system.

b) A uniform solid cylinder has a radius R, mass M, and length L. Calculate its moment of inertia about its central axis, the z axis.

7) A sphere of mass m_1 and a block of mass m_2 are connected by a light cord that passes over a pulley, as shown in the following figure. The radius of the pulley is R, and the mass of the rim is M. The spokes of the pulley have negligible mass. The block slides on a frictionless, horizontal surface. Find an expression for the linear acceleration of the two objects, using the concepts of angular momentum and torque.